Digital Electronics

Notebook

@自由笔记

Title:

Dat

公式化简

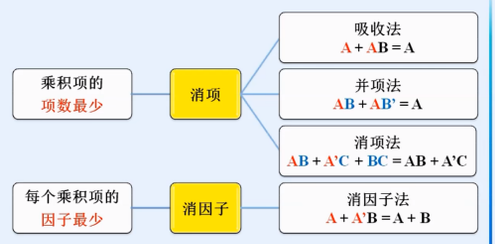
逻辑函数化简 卡诺图

最简与或形式：乘积项项数最少；每个乘积项的因子最少

$Y = A'BC + AB'C + ABC'$

十一、公式化简法

- 1、公式化简法思路



2、吸收法 $A + AB = A$

$Y_1 = ((A'B)' + C)ABD + AD$

$Y_1 = AD((A'B)' + C)B + AD$

$Y_1 = AD$

$Y_2 = AB + ABC' + ABD + AB(C' + D')$

$Y_2 = AB + ABC' + ABD + AB(C' + D')$

$Y_2 = AB$

3、并项法 $AB + AB' = A$

$Y_1 = A(B'CD)' + AB'CD$

$Y_1 = A(B'CD)' + AB'CD$

$Y_1 = A$

$Y_2 = AB' + ACD + A'B' + A'CD$

$Y_2 = A(B' + CD) + A'(B' + CD)$

4、消项法 $AB + A'C + BC = AB + A'C$

$Y_1 = AC + AB' + (B + C)'$

$Y_1 = AC + AB' + B'C'$

$Y_1 = AC + B'C' + AB'$

$Y_1 = AC + B'C'$

$Y_2 = AB'CD' + (AB')'E + A'CD$

$Y_2 = AB'CD' + (AB')'E + A'CD$

$Y_2 = AB'CD' + (AB')'E$

5、消因子法 $A + A'B = A + B$

$Y_1 = B' + ABC$

$Y_1 = B' + BAC$

$Y_1 = B' + AC$

$Y_2 = AB' + B + A'$

$Y_2 = AB' + B + A'$

$Y_2 = B + A + A'B$

$Y_2 = B + A + A'B$

$Y_2 = A + B$

(A) + (B) + (B) = (A) + (B)

6、配项法

利用 $A + A = A$，重复写入某一项

$Y_1 = A'BC' + A'BC + ABC$

$Y_1 = A'BC' + A'BC + A'BC + ABC$

$Y_1 = A'B + BC$

利用 $A + A' = 1$，在某一项上乘以 $A + A'$

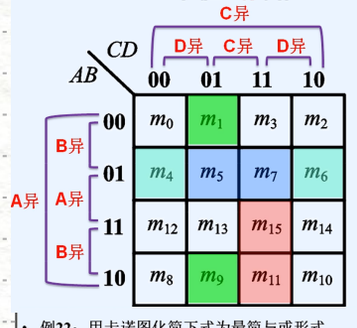
$Y_2 = AB' + A'B + BC' + B'C$

$Y_2 = AB' + A'B(C + C') + BC' + (A + A')B'C$

$Y_2 = AB' + A'BC + A'BC' + BC' + AB'C + A'B'C$

$Y_2 = AB' + AB'C + BC' + A'BC' + A'CB + A'CB'$

$Y_2 = AB' + BC' + A'C$

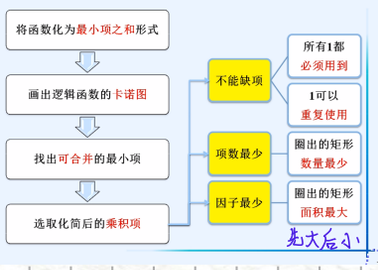


左右相邻

上下相邻

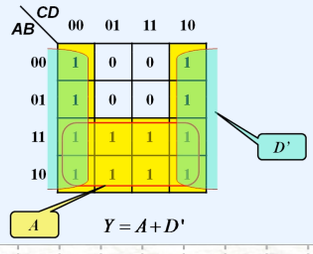
左右 + 上下 = 四角相邻

- 5、卡诺图化简逻辑函数



- 例22：用卡诺图化简下式为最简与或形式

$Y = ABC + ABD + AC'D + C'D' + AB'C + A'CD'$



$Y = A + D'$

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